Light up LED

1. Preface

This lesson will use software to write a program for the LED on the STM32 main control board, controlling the onboard LED to blink.

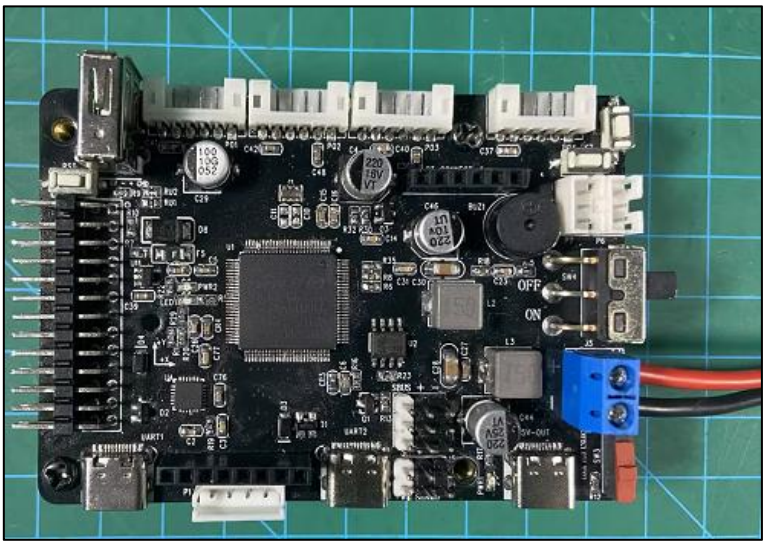
2. Preparation

2.1 Software Preparation

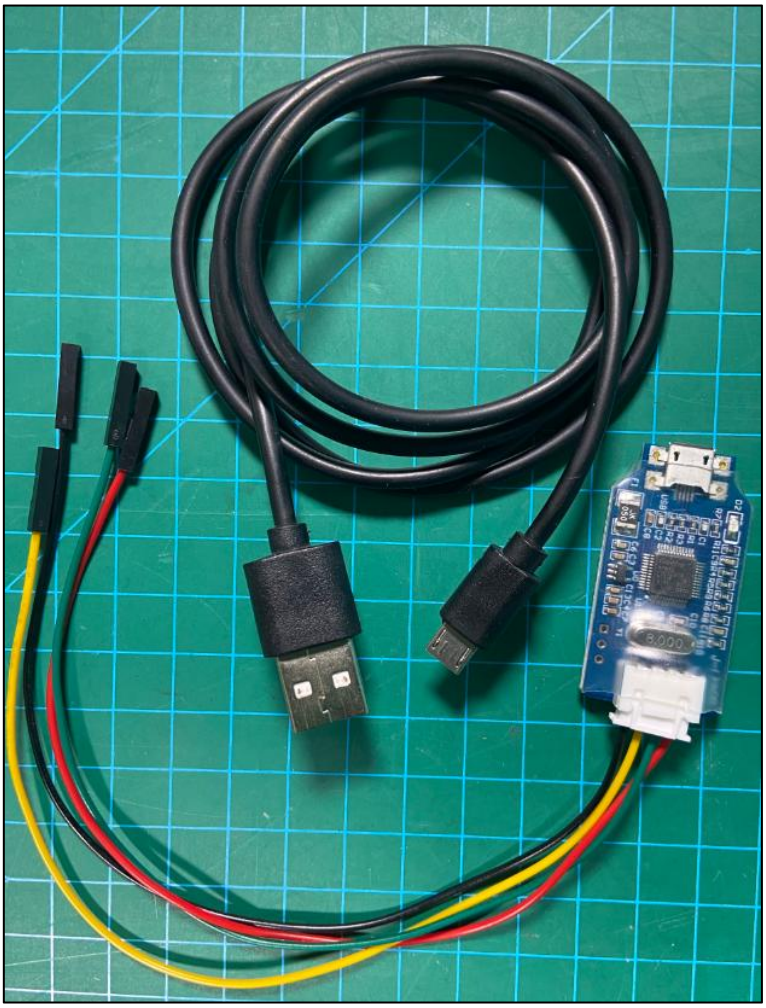
1. Keil5 and related environment configuration (if not installed or configured, installation methods can be found in "**Lesson 1 Development Environment Setup**").
2. J-Link firmware installation (if not installed or configured, installation methods can be found in "**Lesson 1 Development Environment Setup**") for burning.

2.2 Hardware Preparation

1. STM32 Main Controller

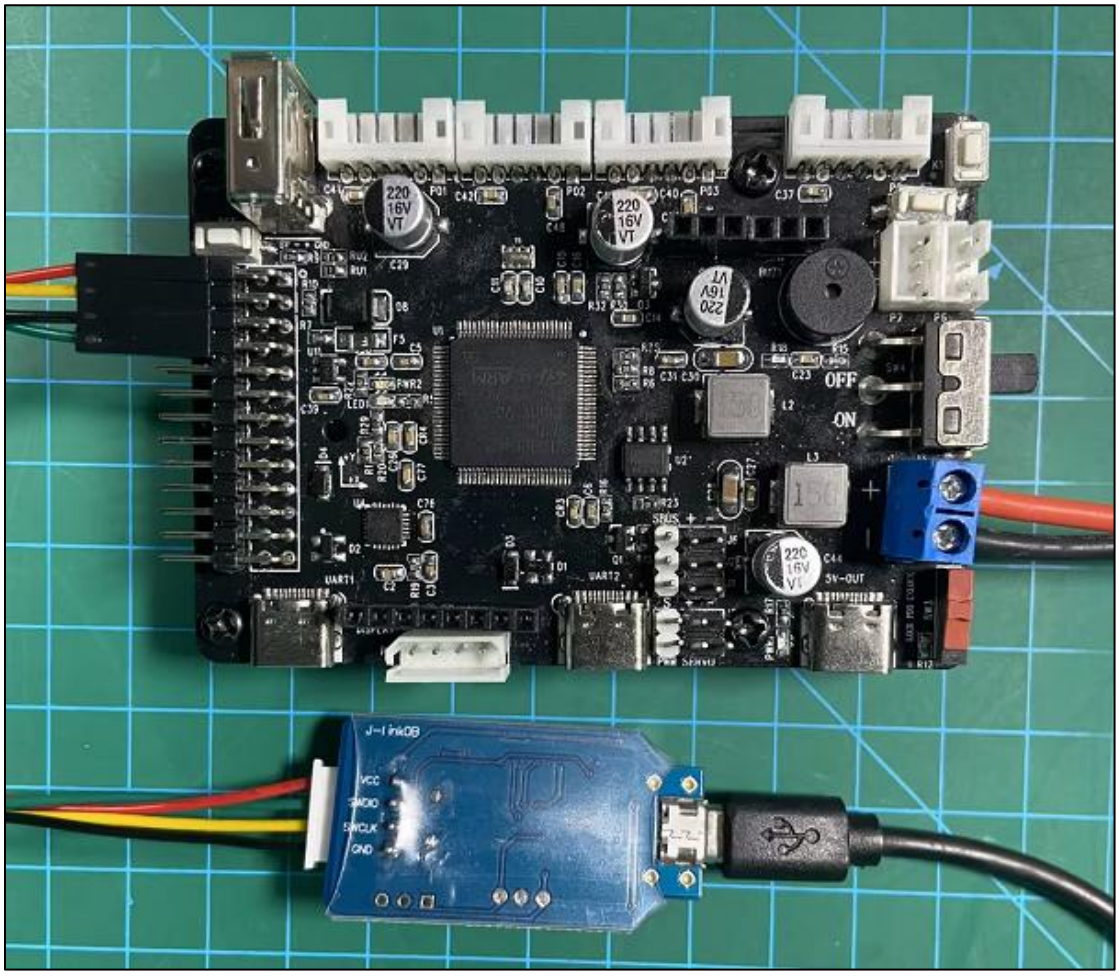


2. The Jlink debugger and a Micro-USB data cable (self-provided) are used for code flashing.



3. Operation Steps

1) After connecting the Jlink debugger and the Micro-USB data cable, insert the DuPont female connector on the other end of the Jlink into the corresponding pin position on the STM32 main control board:



Connect according to the corresponding positions (four wires connected in parallel):

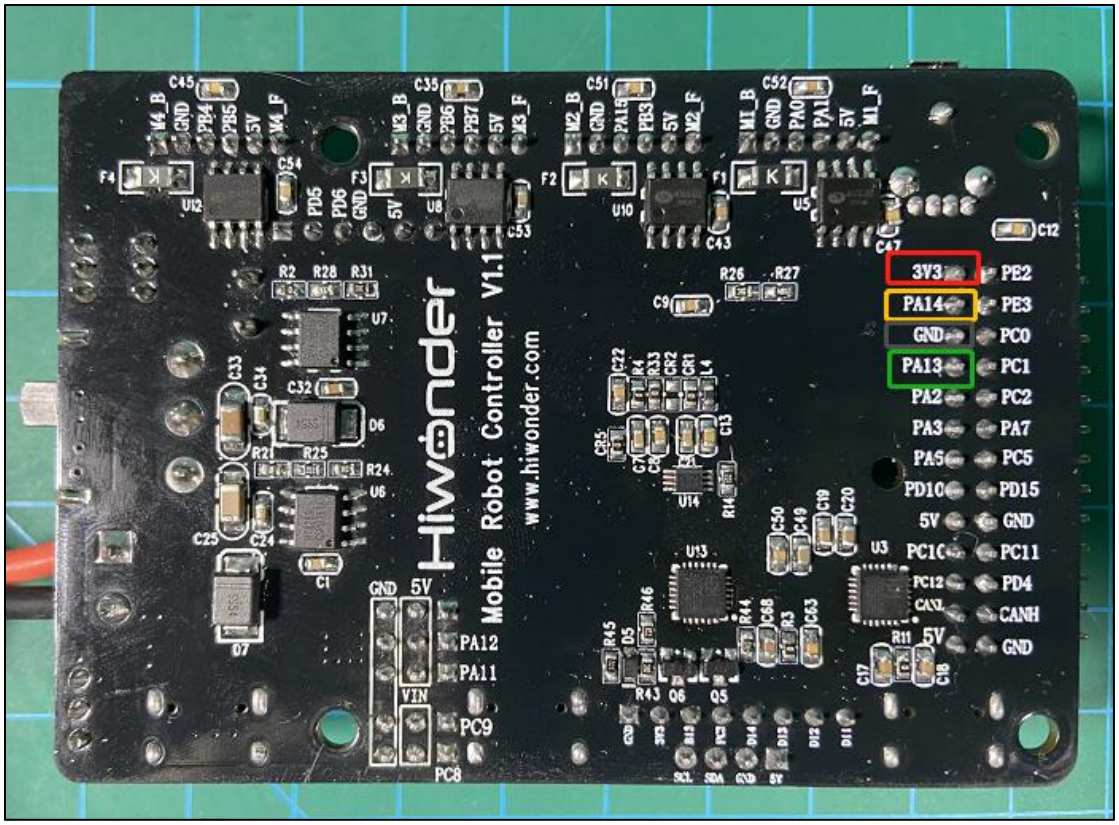
Red VCC connects to the 3V3 pin of the STM32.

Green SWDIO connects to the PA13 pin of the STM32.

Yellow SWCLK connects to the PA14 pin of the STM32.

Black GND connects to the GND pin of the STM32.

The STM32 pin layout is shown in the diagram below:



2) Download the program: For detailed download methods (Jlink download and Type-C serial port download), refer to 'Lesson 3 Project Compilation & Download'".

4. Core Program Analysis

This code is an embedded program based on the STM32 microcontroller, used to control the on/off state of an LED. The program includes several header files, such as main.h, and some library files, such as the GPIO library and the RCC library. In the main function, a series of functions are called to initialize various modules of the STM32 microcontroller, including the system clock, GPIO pins, and the LED.

```
   #include "main.h"int main(void){  // MCU
Configuration--------------------------------------------------------
```

```
HAL_Init(); //initialize MCU  SystemClock_Config(); // Configure Clock
MX_GPIO_Init(); // initialize GPIO pin  while (1) {

 HAL_GPIO_WritePin(GPIOE, GPIO_PIN_10, GPIO_PIN_RESET); // turn off LED
HAL_Delay(2000);

HAL_GPIO_WritePin(GPIOE, GPIO_PIN_10, GPIO_PIN_SET); // turn on LED
HAL_Delay(2000);  }

}
```

In the while loop, the LED blinks every 2 seconds by calling the HAL_GPIO_WritePin function and the HAL_Delay function.

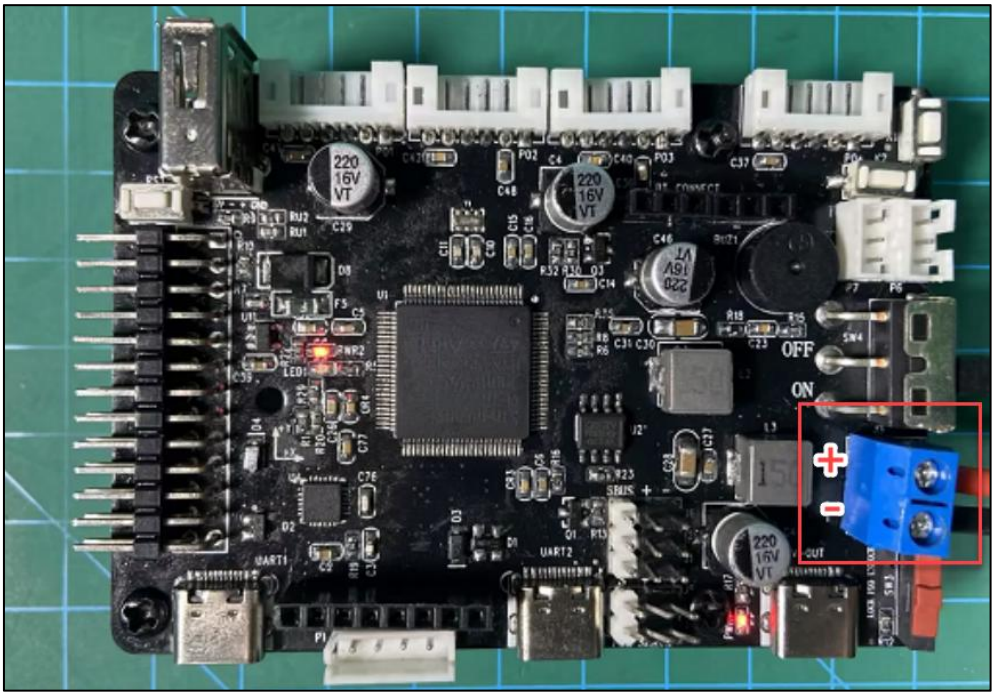
In the main function, the program first calls the HAL_Init function to initialize the MCU, and then calls the SystemClock_Config function and the MX_GPIO_Init function to configure the system clock and GPIO pins, respectively. These functions are generated by STM32 CubeMX to initialize various functional modules of the STM32 microcontroller.

In the while loop, the program calls the HAL_GPIO_WritePin function and the HAL_Delay function to make the LED blink every 2 seconds. The HAL_GPIO_WritePin function controls the voltage level of the GPIO pin, turning the LED off and on by passing the GPIO_PIN_RESET and GPIO_PIN_SET parameters, respectively. The HAL_Delay function creates a delay, pausing the program for a specified time before executing the next statement, which in this case, controls the blinking interval of the LED.

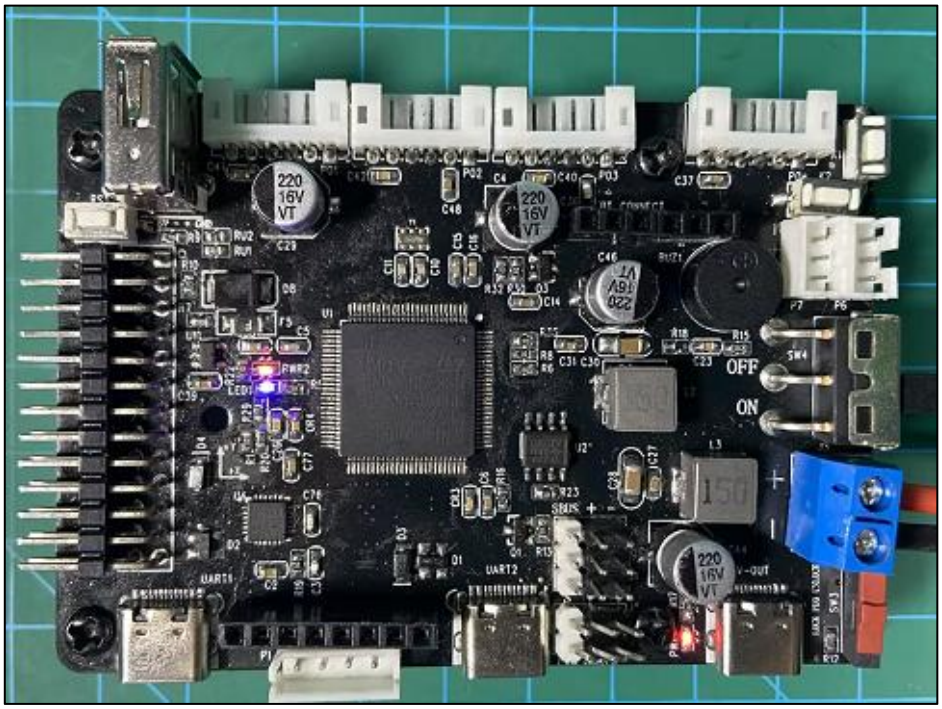
5. Program Outcome

After flashing the program, you can disconnect the Jlink programming cable and provide the STM32 with a 7.4V or 11.4V battery power supply. As shown in the two diagrams below, connect the '＋' terminal to the red positive terminal of

the lithium battery and the '—' terminal to the negative terminal of the lithium battery.



Effect Diagram 1 (LED Light Off)



Effect Diagram 2 (LED Light On)